Solution Requirements (Functional & Non-functional)

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement(Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input	Enter Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall values Validate input values before prediction
FR-2	Crop Recommendation	Process user input using the trained ML model,Display the recommended crop instantly
FR-3	Machine learning Prediction	Load the trained Random Forest model (.pkl) Generate crop prediction based on input parameters
FR-4	Result Display	Show predicted crop with a user-friendly interface,Allow users to make multiple predictions without restarting the application
FR-5	Data Processing	Preprocess user inputs before prediction,Handle invalid or missing input values
FR-6	Model Management	Store and load the trained model using Pickle, enable future model updates using new datasets

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No	Non-Functional Requirement	Description
NFR-1	**Usability**	The web application should have a simple, responsive, and user-friendly interface for farmers.
NFR-2	**Security**	Validate all user inputs and protect the application from invalid or malicious data.
NFR-3	**Reliability**	The system should consistently provide accurate crop recommendations using the trained model.
NFR-4	**Performance**	Crop prediction should be generated within 2–3 seconds after input submission.
NFR-5	**Availability**	The Flask application should be available whenever the server is running and support continuous usage.
NFR-6	**Scalability**	The system should support future enhancements such as fertilizer recommendation, weather forecasting, disease detection, and cloud deployment.